

BBA DATA SCIENCE & AI • AIM

Agentic AI

Architecture, Principles & Business Applications

Logistics · Digital Marketing · Business Analytics · Strategic HR · Fintech



From autonomous reasoning systems to real-world enterprise impact

What We Will Cover

PART ONE

The Architecture of Agentic AI

- What makes AI "agentic"
- Core architectural components
- Guiding design principles
- Single vs. multi-agent systems
- Vertical, horizontal & hybrid models
- Reactive, deliberative & cognitive frameworks

PART TWO

Agentic AI in Business

- Logistics & supply chain management
- Digital marketing
- Business analytics
- Strategic human resource management
- Fintech
- Challenges & key takeaways

What Is Agentic AI?

Agentic AI systems act on their own and pursue goals in changing environments. Unlike traditional AI, which depends on step-by-step human instruction, an agentic system can observe its environment, reason about what it sees, learn from experience, and decide what to do next — with minimal human intervention.

Because it decides its own path to a goal, one agentic system can handle demand shifts, customer queries, or supply disruptions that a rule-based tool would simply fail on.

THE AGENTIC LOOP



Perceive

Sense the environment



Reason

Evaluate against goals



Act

Execute autonomously



Learn

Adapt from feedback

Traditional AI vs. Agentic AI

	Traditional AI	Agentic AI
Human role	Directs every step	Sets the goal; agent decides the steps
Operation	Executes a fixed instruction	Observes, plans and adapts continuously
Decision making	Rule-bound, single response	Multi-step reasoning toward an objective
Memory & learning	Typically stateless per query	Learns from feedback over time
Best suited for	Well-defined, narrow tasks	Open-ended, dynamic business problems

The Four Core Components

Every agentic system is built from four interlocking components that let it sense, decide, act and improve on its own.

01



Perception

Collects information from its surroundings — sensors, data streams, external databases and APIs.

02



Cognitive Layer

Analyzes the data and chooses the best action using rules, ML models or reinforcement learning.

03



Action & Execution

Carries out the chosen decision — controlling robotics or automating software workflows.

04



Learning & Adaptation

Improves over time via supervised, unsupervised or reinforcement learning.

Perception



How an agent collects information from its surroundings — using images, sound, text or sensor data to understand its environment and recognize events that need a response.



Sensors

- Cameras & microphones
- Motion detectors
- Temperature & location sensors
- Specialized environment monitors



Data Integration

- IoT device streams
- External APIs
- Historical datasets
- A unified, comprehensive view of the situation

The Cognitive Layer



After understanding its environment, the agent must analyze the data and decide the best action — assessing the situation, weighing outcomes, and choosing a path toward its goal.



Rule-Based Systems

Simple systems that follow predefined rules to make decisions.



Machine Learning Models

Advanced systems using statistical techniques to learn patterns and predict.



Reinforcement Learning

Learns through trial and error via rewards or penalties on its actions.

Action & Execution



Once the agent chooses an action, this component carries it out in the environment — from physical control to fully automated digital workflows.



Robotics

Controls robotic systems in physical environments — assembly, navigation and interaction with humans (e.g. robotic arms, self-driving vehicles).



Software Automation

Controls virtual systems to automate business decision-making, customer service chatbots, or IT systems management.

Learning & Adaptation



Agentic systems adapt and improve over time by learning from past experience — enabling them to handle new situations they were never explicitly programmed for.



Supervised Learning

The agent is trained on labeled data to make predictions or classifications.



Unsupervised Learning

Identifies patterns in unlabeled data without predefined categories.



Reinforcement Learning

Learns through trial and error, improving via rewards or penalties over time.

Principles of Agentic AI Design



Autonomy

Works independently within set limits, reducing human involvement while following ethical and safety guidelines.



Goal-Directed Behavior

Focuses on clear objectives that guide perception, reasoning and planning — set by users or inferred from context.



Adaptability

Improves over time by learning from feedback, using methods like online or meta-learning.



Modularity

A modular design lets components be built, tested and updated independently — improving scalability.



Transparency

Explains its reasoning and actions to build trust — critical in domains like finance and healthcare.

Single-Agent vs. Multi-Agent Systems



Single-Agent Architecture

A single AI system that functions independently — making decisions and taking actions on its own, without coordinating with other agents.

Best for: focused, well-scoped tasks such as a single chatbot or a single monitoring agent.



Multi-Agent Architecture

Multiple AI systems interact, collaborate and coordinate their actions to achieve common goals — each may specialize in a domain.

Three sub-types: Vertical, Horizontal & Hybrid — explored next.

Vertical AI Architecture



Key Features

- Clearly defined hierarchy
- Centralized communication
- Agents report to a leader



Strengths

- Ideal for sequential workflows
- Clear accountability — leader aligns objectives



Weaknesses

- Leader reliance can slow progress
- Single point of failure



Use Cases

- Multistep workflow automation
- Document generation with leader oversight

Horizontal AI Architecture



Key Features

- Peer collaboration model
- Distributed, decentralized decisions
- Agents share resources & ideas freely



Strengths

- Fosters dynamic problem solving
- Parallel processing of tasks



Weaknesses

- Coordination challenges
- Slower decisions from deliberation



Use Cases

- Brainstorming diverse ideas
- Complex, interdisciplinary problem solving

Hybrid AI Architecture



Key Features

- Combines structured leadership + flexibility
- Leadership shifts by task phase
- Peers engaged openly by leaders



Strengths

- Versatile — combines both models
- Handles structure + creativity together



Weaknesses

- Complex to balance roles
- More demanding resource management



Use Cases

- Strategic planning & team projects
- Dynamic processes needing both modes

Agentic Frameworks

A structured system defining how agents interact, decide and execute — integrating planning, memory, tool use and communication.



Reactive Architectures

Link immediate environmental input directly to responses. Agents act on instinct — no memory or planning, and cannot learn from the past or anticipate the future.



Deliberative Architectures

Agents reason, strategize and build internal representations of the world — assessing surroundings and considering outcomes before deciding.



Cognitive Architectures

Simulate human-like intelligence, combining perception, reasoning, learning and decision-making to handle complex tasks adaptively.

PART TWO

Agentic AI in Business

Logistics & Supply Chain · Digital Marketing · Business Analytics · Strategic HR · Fintech

Why Business Is Turning to Agentic AI

The same components that let an agent perceive, reason, act and learn translate directly into business value — across every function.



24/7 Autonomous Operation

Agents monitor, decide and act continuously — without waiting on human bandwidth.



Faster, Data-Driven Decisions

Multi-step reasoning turns raw business data into action in real time.



Adapts to Change

Learns from outcomes — demand spikes, market shifts, customer behaviour.



Scales Across Functions

Modular agents can be deployed department by department, then connected.

Agentic AI in Logistics & Supply Chain



Supply chains are dynamic, multi-party environments — exactly where autonomous, goal-directed agents outperform static, rule-based software.

AGENTIC COMPONENT → BUSINESS FUNCTION

Perception	→ IoT sensors, GPS & warehouse systems feed live shipment, stock and traffic data
Cognitive Layer	→ Agents forecast demand and evaluate routing, sourcing and inventory trade-offs
Action & Execution	→ Autonomously reroutes shipments, reorders stock, schedules warehouse robots
Learning & Adaptation	→ Improves forecasts each cycle by learning from delays, demand shifts, disruptions

Logistics & Supply Chain — Key Use Cases



Autonomous Route Optimization

Agents continuously re-plan delivery routes around traffic, weather and fuel cost, without manual dispatch.



Demand Forecasting & Inventory

Predicts SKU-level demand and auto-triggers replenishment to avoid stockouts and overstock.



Warehouse & Fleet Automation

Coordinates robotic picking, packing and fleet scheduling as a multi-agent team.



Disruption Response

Detects delays or supplier risk early and autonomously re-sources or re-routes to protect delivery SLAs.

Agentic AI in Digital Marketing



Marketing generates constant behavioural data and requires rapid, personalized response — a natural fit for perceive-decide-act-learn agents.

AGENTIC COMPONENT → BUSINESS FUNCTION

Perception	→ Tracks browsing, engagement, campaign and social signals in real time
Cognitive Layer	→ Evaluates audience segments, channel mix and content-market fit against goals
Action & Execution	→ Launches campaigns, personalizes content, adjusts ad bids autonomously
Learning & Adaptation	→ Refines targeting and creative continuously from conversion feedback

Digital Marketing — Key Use Cases



Autonomous Campaign Agents

Plan, launch and adjust multi-channel campaigns against a stated goal — e.g. "maximize qualified leads."



Hyper-Personalization

Tailors content, offers and product recommendations per-visitor, in real time.



Ad-Spend & Bid Optimization

Continuously reallocates budget across channels toward the best-performing segments.



Conversational Commerce Agents

Handle discovery-to-purchase conversations across chat and social without a human in the loop.

Agentic AI in Business Analytics



Agentic analytics systems go beyond dashboards — they investigate anomalies, form hypotheses, and surface decisions on their own.



Autonomous Insight Discovery

Continuously mines data for meaningful patterns and surfaces them without a query being written.



Anomaly & Risk Detection

Flags irregular metrics — fraud, churn risk, quality drift — the moment they emerge.



Self-Service Data Pipelines

Agents clean, join and prepare data automatically as new sources appear.



Decision-Support Agents

Translate findings into recommended actions tied to business KPIs, not just charts.

Agentic AI in Strategic HR Management



From hiring to retention, HR generates continuous signal about people — agentic systems help HR move from administration to strategy.



Intelligent Talent Sourcing

Screens, ranks and shortlists candidates against role requirements autonomously.



Onboarding & Employee Support

Agent-driven onboarding journeys and 24/7 HR query resolution for employees.



Attrition Risk Prediction

Flags flight-risk employees early using engagement, performance and sentiment data.



Workforce Planning

Continuously models skill gaps and staffing needs against business strategy.

Agentic AI in Fintech



Finance is fast, rule-heavy, and rich in data — ideal terrain for agents that must act within strict guardrails and explain themselves.



Real-Time Fraud Detection

Monitors transactions continuously and autonomously blocks or escalates suspicious activity.



Algorithmic Trading Agents

Perceive market signals and execute trades within defined risk limits.



Autonomous Credit Underwriting

Assesses risk and approves or routes loan applications using live financial data.



Personal Finance Assistants

Give customers proactive budgeting, savings and investment guidance in real time.

KEY TAKEAWAYS

Agentic AI Turns Autonomy Into Advantage



Same architecture, every industry

Perception, cognition, action and learning apply identically to logistics, marketing, analytics, HR and fintech.



Choose the right model

Single-agent for narrow tasks; vertical, horizontal or hybrid multi-agent for complex, cross-functional work.



Transparency is non-negotiable

In finance, HR and healthcare especially, agents must be able to explain their reasoning.



Autonomy needs guardrails

Clear goals, ethical limits and human oversight keep agentic systems safe at scale.

Thank You